

Self Trigger

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

The function of Self Trigger is described. This function can set 2 areas from effective pixels which are Sensing Area and Capturing Area. And when the number of pixels which are over the level threshold is over the count threshold, it is as trigger, the image of Capturing Area is captured.

3. Features

◆ Self Trigger mode

One Sensing Area and one Capturing Area can be set.
 Set level threshold of each pixel in sensing area.
 Set count threshold for the number of pixels over the level threshold.
 Turn on and turn off the Sensing frame output.
 Output alternately the Sensing frame and Capturing frame. (Test function)

◆ Frame identifier

Frame information line (In case of SLVS output)
 Embedded data line (In case of SLVS-EC output or CSI-2 serial data output)

◆ Support mode

Normal mode (Sensor Master mode only)

◆ Support readout drive mode

All-pixel scan mode
 Vertical / Horizontal 1/2 Subsampling mode
 2 × 2 FD binning mode (The product with a color filter is excluded)

◆ The monitor output for Exposure period

◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

◆ Gain adjust function per each frame

Each gain of Sensing Area and Capturing Area can be set.

Contents

1.	Product type -----	1
2.	Description -----	1
3.	Features -----	2
	Contents-----	3
4.	Register Map -----	4
5.	Register Communication Timing and Reflection Timing-----	14
6.	Description of Self Trigger Function -----	15
7.	Self Trigger Setting-----	16
7.1.	ROI Setting -----	16
7.2.	Exposure Time Setting -----	16
7.3.	Gain Setting -----	16
7.4.	Threshold Setting -----	17
8.	Supporting Function of Setting Threshold-----	20
8.1.	Sensing Frame Output Setting -----	20
8.2.	How to Confirm Number of Count -----	20
8.3.	Self Trigger Forcible Switching Function-----	20
9.	Exposure Timing and Output Timing -----	21
10.	The monitor output for Exposure period -----	22
10.1.	List of registers -----	22
11.	Frame Information-----	23
11.1.	Frame information line-----	23
11.2.	Output format (Frame information line)-----	23
11.3.	Embedded data line-----	24
11.4.	Output format (Embedded data line)-----	24
12.	Mode Transitions -----	26
12.1.	Mode Transition -----	26
13.	Note-----	27
	Revision History-----	28

4. Register Map

The register map for Dual Trigger is shown below.

Please refer to the product specification for register setup other than those listed.

Registers corresponding to Chip ID = 02h in Write mode. (I²C:30**h)

Address		bit	Register Name	Description	Default value after reset						Reflection timing			
4-wire	I ² C				By register			By address						
D4h	30D4h	0	VMAX [23:0]	LSB	0012 ECh ^{*1}	000C D4h ^{*2}	0008 AAh ^{*3}	ECh *1	D4h *2	AAh *3	V			
		1												
		2												
		3												
		4												
		5												
		6												
		7												
D5h	30D5h	0		When sensor master mode vertical span setting for Capturing frame. (Number of operation lines count from 1)				0012 ECh ^{*1}	000C D4h ^{*2}	0008 AAh ^{*3}		12h *1	0Ch *2	08h *3
		1												
		2												
		3												
		4												
		5												
		6												
		7												
D6h	30D6h	0		MSB				0012 ECh ^{*1}	000C D4h ^{*2}	0008 AAh ^{*3}		12h *1	0Ch *2	08h *3
		1												
		2												
		3												
		4												
		5												
		6												
		7												

^{*1} IMX530 series and IMX540 series.

^{*2} IMX535 series, IMX545 series and IMX565 series.

^{*3} IMX548 series and IMX568 series.

Registers corresponding to Chip ID = 03h in Write mode. (I²C:31**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
04h	3104h	0	FID0_ROIH1ON [0]	The horizontal setting of Sensing Area Fixed to "1"	0	00h	V
		1	FID0_ROIV1ON [0]	The horizontal setting of Sensing Area Fixed to "1"	0		I
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
06h	3106h	0	FID1_ROIH1ON [0]	The horizontal setting of Capturing Area Fixed to "1"	0	00h	V
		1	FID1_ROIV1ON [0]	The horizontal setting of Capturing Area Fixed to "1"	0		I
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
20h	3120h	0	FID0_ROIPH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
		7					
21h	3121h	0	FID0_ROIPH1 [12:0]	Designation of horizontal cropping position on Sensing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
22h	3122h	0	FID0_ROIPV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
		7					
23h	3123h	0	FID0_ROIPV1 [12:0]	Designation of vertical cropping position on Sensing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	I
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
24h	3124h	0	FID0_ROIWH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
25h	3125h	7	FID0_ROIWH1 [12:0]	Designation of horizontal cropping size on Sensing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
26h	3126h	6	FID0_ROIWH1 [12:0]	Designation of horizontal cropping size on Sensing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		7					
		0					
		1					
		2					
		3					
		4					
27h	3127h	5	FID0_ROIWH1 [12:0]	Designation of horizontal cropping size on Sensing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		6					
		7					
		0					
		1					
		2					
		3					
60h	3160h	4	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
61h	3161h	3	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		4					
		5					
		6					
		7					
		0					
		1					
61h	3161h	2	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		3					
		4					
		5					
		6					
		7					
		0					
61h	3161h	1	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		2					
		3					
		4					
		5					
		6					
		7					
61h	3161h	0	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
61h	3161h	7	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
61h	3161h	6	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		7					
		0					
		1					
		2					
		3					
		4					
61h	3161h	5	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		6					
		7					
		0					
		1					
		2					
		3					
61h	3161h	4	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
61h	3161h	3	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		4					
		5					
		6					
		7					
		0					
		1					
61h	3161h	2	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		3					
		4					
		5					
		6					
		7					
		0					
61h	3161h	1	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		2					
		3					
		4					
		5					
		6					
		7					
61h	3161h	0	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
61h	3161h	7	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
61h	3161h	6	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		7					
		0					
		1					
		2					
		3					
		4					
61h	3161h	5	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		6					
		7					
		0					
		1					
		2					
		3					
61h	3161h	4	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
61h	3161h	3	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		4					
		5					
		6					
		7					
		0					
		1					
61h	3161h	2	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		3					
		4					
		5					
		6					
		7					
		0					
61h	3161h	1	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		2					
		3					
		4					
		5					
		6					
		7					
61h	3161h	0	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
61h	3161h	7	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
61h	3161h	6	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		7					
		0					
		1					
		2					
		3					
		4					
61h	3161h	5	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		6					
		7					
		0					
		1					
		2					
		3					
61h	3161h	4	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
61h	3161h	3	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		4					
		5					
		6					
		7					
		0					
		1					
61h	3161h	2	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		3					
		4					
		5					
		6					
		7					
		0					
61h	3161h	1	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		2					
		3					
		4					
		5					
		6					
		7					
61h	3161h	0	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
61h	3161h	7	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
61h	3161h	6	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		7					
		0					
		1					
		2					
		3					
		4					
61h	3161h	5	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		6					
		7					
		0					
		1					
		2					
		3					
61h	3161h	4	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
61h	3161h	3	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		4					
		5					
		6					
		7					
		0					
		1					
61h	3161h	2	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		3					
		4					
		5					
		6					
		7					
		0					
61h	3161h	1	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		2					
		3					
		4					
		5					
		6					
		7					
61h	3161h	0	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
61h	3161h	7	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
61h	3161h	6	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		7					
		0					
		1					
		2					
		3					
		4					
61h	3161h	5	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		6					
		7					
		0					
		1					
		2					
		3					
61h	3161h	4	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
61h	3161h	3	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		4					
		5					
		6					
		7					
		0					
		1					
61h	3161h	2	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		3					
		4					
		5					
		6					
		7					
		0					
61h	3161h	1	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		2					
		3					
		4					
		5					
		6					
		7					
61h	3161h	0	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
61h	3161h	7	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
61h	3161h	6	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		7					
		0					
		1					
		2					
		3					
		4					
61h	3161h	5	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		6					
		7					
		0					
		1					
		2					
		3					
61h	3161h	4	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
61h	3161h	3	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		4					
		5					
		6					
		7					
		0					
		1					
61h	3161h	2	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		3					
		4					
		5					
		6					
		7					
		0					
61h	3161h	1	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		2					
		3					
		4					
		5					
		6					
		7					
61h	3161h	0	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
61h	3161h	7	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		0					
		1					
		2					
		3					
		4					
		5					
61h	3161h	6	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		7					
		0					
		1					
		2					
		3					
		4					
61h	3161h	5	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		6					
		7					
		0					
		1					
		2					
		3					
61h	3161h	4	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
61h	3161h	3	FID1_ROIPH1 [12:0]	Designation of horizontal cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}	0000h	0	

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
62h	3162h	0	FID1_ROIPV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
63h	3163h	7		Designation of vertical cropping position on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}			
		0	FID1_ROIWH1 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0	0		—
64h	3164h	7		Fixed to 0	0		—
		0	FID1_ROIWH1 [12:0]		0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of horizontal cropping size on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}			
65h	3165h	7		MSB			
		0	FID1_ROIWH1 [12:0]		0000h	00h	I
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0	0		—
66h	3166h	7		Fixed to 0	0		—
		0	FID1_ROIWH1 [12:0]		0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Designation of vertical cropping size on Capturing Area *Set the value of multiple of 16 ^{*3} *Set the value of multiple of 8 ^{*4}			
67h	3167h	7		MSB			
		0	FID1_ROIWH1 [12:0]		0000h	00h	I
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0	0		—
67h	3167h	7		Fixed to 0	0		—
		0	FID1_ROIWH1 [12:0]		0000h	00h	I
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0	0		—
67h	3167h	7		Fixed to 0	0		—

*3 IMX530 series and IMX535 series.

*4 IMX540 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

Registers corresponding to Chip ID = 04h in Write mode. (I²C:32**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing																				
4-wire	I ² C				By register	By address																					
40h	3240h	0	SHS [23:0]	LSB	000048h*1	000070h*2	48h*1	70h*2	V																		
		1																									
		2																									
		3																									
		4																									
		5																									
		6																									
7																											
41h	3241h	0					Storage time adjustment: Sensing Frame Designated in line unit	000048h*1		000070h*2	00h	V															
		1																									
		2																									
		3																									
		4																									
		5																									
		6																									
7																											
42h	3242h	0											000048h*1	000070h*2	00h	V											
		1																									
		2																									
		3																									
		4																									
		5																									
		6																									
7																											
		0		MSB																							
		1																									
		2																									
		3																									
		4																									
		5																									
		6																									
7																											
44h	3244h	0																				48h*1	70h*2				
		1																									
		2																									
		3																									
		4																									
		5																									
		6																									
7																											
45h	3245h	0									SHS2 [23:0]											LSB	000048h*1		000070h*2	00h	V
		1																									
		2																									
		3																									
		4																									
		5																									
		6																									
7																											
46h	3246h	0		MSB					00h																		
		1																									
		2																									
		3																									
		4																									
		5																									
		6																									
7																											

^{*1} IMX530 series and IMX540 series.

^{*2} IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

Registers corresponding to Chip ID = 06h in Write mode. (I²C:34**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing		
4-wire	I ² C				By register	By address			
00h	3400h	0	TRIGMODE [2:0]	Global shutter mode setting 0h: Normal mode 3h: Self Trigger mode Others: Setting prohibited	0h	00h	I ^{*1}		
		1							
		2							
		3	TRIGTIMING [1:0]	Trigger mode setting 0h: Normal mode Other setting is prohibited.	0h		I ^{*1}		
		4							
		5		Fixed to 0				0	—
		6		Fixed to 0				0	—
		7	Fixed to 0	0	—				
14h	3414h	0	VMAX_STRIG [23:0]	LSB 					

^{*1} Please set Self trigger mode in the state of the standby.

^{*2} IMX530 series and IMX540 series.

^{*3} IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
1Ah	341Ah	0	SELFTRIG_TH_L [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6		Self Trigger Level threshold L side			
		7					
1Bh	341Bh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
1Ch	341Ch	0		LSB		00h	
		1					
		2					
		3					
		4					
		5					
		6					
		7					
1Dh	341Dh	0	SELFTRIG_CNT_TH_H [23:0]		000000h	00h	I
		1					
		2					
		3					
		4		Self Trigger Count threshold H side			
		5					
		6					
		7					
1Eh	341Eh	0				00h	
		1					
		2					
		3					
		4					
		5					
		6					
		7		MSB			

Address		bit	Register Name	Description	Default value after reset		Reflection timing	
4-wire	I ² C				By register	By address		
20h	3420h	0	SELFTRIG_CNT_TH_L [23:0]	LSB	000000h	00h	I	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
21h	3421h	0		Self Trigger Count threshold L side		000000h		00h
		1						
		2						
		3						
		4						
		5						
		6						
		7						
22h	3422h	0						00h
		1						
		2						
		3						
		4						
		5						
		6						
		7						
24h	3424h	0	SELFTRIG_MONITOR_T EST [0]	Sensing frame output setting 0: Without output 1: With output	0		00h	I
		1	SELF_TRG_FORCE_ON [0]	Self Trigger forcible switching 0: Normal mode 1: Forced mode (Sensing frame and Capturing frame is switched by each frame.)	0			I
		2		Fixed to 0	0			—
		3		Fixed to 0	0			—
		4		Fixed to 0	0			—
		5		Fixed to 0	0			—
		6		Fixed to 0	0			—
		7		Fixed to 0	0			—

Address		bit	Register Name	Description	Default value after reset		Reflection timing										
4-wire	I ² C				By register	By address											
28h	3428h	0	SELFTRIG_CNT_H [23:0]	Read out the number of detected pixel H side	Undefined	Undefined	I										
		1															
		2															
		3															
		4															
		5															
		6															
		7															
29h	3429h	0						SELFTRIG_CNT_H [23:0]	Read out the number of detected pixel H side	Undefined	Undefined	I					
		1															
		2															
		3															
		4															
		5															
		6															
		7															
2Ah	342Ah	0											SELFTRIG_CNT_H [23:0]	Read out the number of detected pixel H side	Undefined	Undefined	I
		1															
		2															
		3															
		4															
		5															
		6															
		7															
2Ch	342Ch	0	SELFTRIG_CNT_L [23:0]	Read out the number of detected pixel L side	Undefined	Undefined	I										
		1															
		2															
		3															
		4															
		5															
		6															
		7															
2Dh	342Dh	0						SELFTRIG_CNT_L [23:0]	Read out the number of detected pixel L side	Undefined	Undefined	I					
		1															
		2															
		3															
		4															
		5															
		6															
		7															
2Eh	342Eh	0											SELFTRIG_CNT_L [23:0]	Read out the number of detected pixel L side	Undefined	Undefined	I
		1															
		2															
		3															
		4															
		5															
		6															
		7															

Registers corresponding to Chip ID = 07h in Write mode. (I²C:35**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
14h	3514h	0	GAIN [8:0]	LSB	000h	00h	V
		1		Gain setting for Sensing frame			
		2					
		3					
		4					
		5					
		6					
		7					
15h	3515h	0		MSB	0	00h	—
		1		Fixed to 0			
		2		Fixed to 0			
		3		Fixed to 0			
		4		Fixed to 0			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
16h	3516h	0	GAIN1 [8:0]	LSB	000h	00h	V
		1		Gain setting for Capturing frame			
		2					
		3					
		4					
		5					
		6					
		7					
17h	3517h	0		MSB	0	00h	—
		1		Fixed to 0			
		2		Fixed to 0			
		3		Fixed to 0			
		4		Fixed to 0			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

5. Register Communication Timing and Reflection Timing

In this mode, Communication timing and Reflection Timing is same as Global Shutter (Normal mode) which described at Register Communication Timing of Datasheet.

FRAMOS Technologies Inc.
Internal use only

6. Description of Self Trigger Function

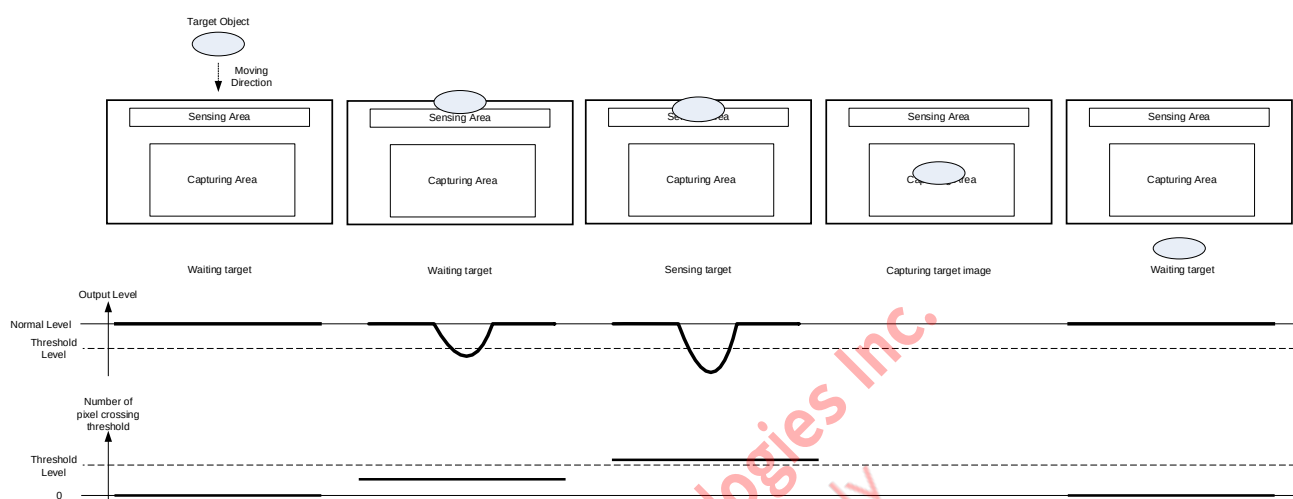
By setting Sensing Area and Capturing Area in effective pixel and counting pixels which are over Level threshold, this function outputs Capturing Area Image when the number of pixels is over the Count threshold.

By setting Sensing Area small, the frame rate is faster, and the Sensing speed is faster.

Level threshold and Count threshold can set high side (bright side) low side (dark side).

Gain is set for each area.

This function supports Master mode only.



Function Image of Sensing Low Level Threshold

7. Self Trigger Setting

7.1. ROI Setting

Sensing Area and Capturing Area can be set independently. Restriction of each area setting is same as the case which one area is set at ROI mode. *1

2 areas can be overlapped.

By setting vertical width of Sensing Area narrow, the sensing speed is faster.

About each frame rate, VMAX (Chip ID = 02h, Address = D4h, D5h, D6h) and VMAX_STRIG (Chip ID = 06h, Address = 14h, 15h, 16h) of Sensing frame and Capturing frame, refer to ROI mode of Datasheet. *2

*1 This function supports All pixel scan mode only.

*2 Restriction of exposure time is added. Refer to section of Exposure Time Setting.

Output image of Sensing Area and Capturing Area is same as ROI mode of Datasheet.

7.2. Exposure Time Setting

Exposure time is set as below.

Set registers as below

Chip ID	Address	Bit	register name	Note
02h	D4h	[7:0]	VMAX [23:0]	When sensor master mode vertical span setting for Capturing frame. (Number of operation lines count from 1)
	D5h	[7:0]		
	D6h	[7:0]		
04h	40h	[7:0]	SHS [23:0]	Storage time adjustment: Sensing Frame Designated in line unit
	41h	[7:0]		
	42h	[7:0]		
	44h	[7:0]	SHS2 [23:0]	Storage time adjustment: Capturing Frame Designated in line unit
	45h	[7:0]		
	46h	[7:0]		
06h	14h	[7:0]	VMAX_STRIG [23:0]	When sensor master mode vertical span setting for Sensing frame. (Number of operation lines count from 1)
	15h	[7:0]		
	16h	[7:0]		

At setting exposure time, set the register satisfied the equation.

$$VMAX_STRIG - SHS = VMAX - SHS2$$

7.3. Gain Setting

Set registers as below

Chip ID	Address	Bit	register name	Note
07h	14h	[7:0]	GAIN	Gain setting for Sensing frame
	15h	[0]		
	16h	[7:0]	GAIN1	Gain setting for Capturing frame
	17h	[0]		

7.4. Threshold Setting

There are 2 thresholds, Level threshold and Count threshold. And Each threshold has H side and L side.

Set registers as below

Chip ID	Address	Bit	register name	Note
06h	18h	[7:0]	SELFTRIG_TH_H [12:0]	Self Trigger Level threshold H side
	19h	[4:0]		
	1Ah	[7:0]	SELFTRIG_TH_L [12:0]	Self Trigger Level threshold L side
	1Bh	[4:0]		
	1Ch	[7:0]	SELFTRIG_CNT_TH_H [23:0]	Self Trigger Count threshold H side
	1Dh	[7:0]		
	1Eh	[7:0]		
	20h	[7:0]	SELFTRIG_CNT_TH_L [23:0]	Self Trigger Count threshold L side
	21h	[7:0]		
	22h	[7:0]		

FRAMOS Technologies Inc.
Internal use only

SELFTRIG_TH_H and SELFTRIG_TH_L are Level thresholds which is to judge the target in each pixel.

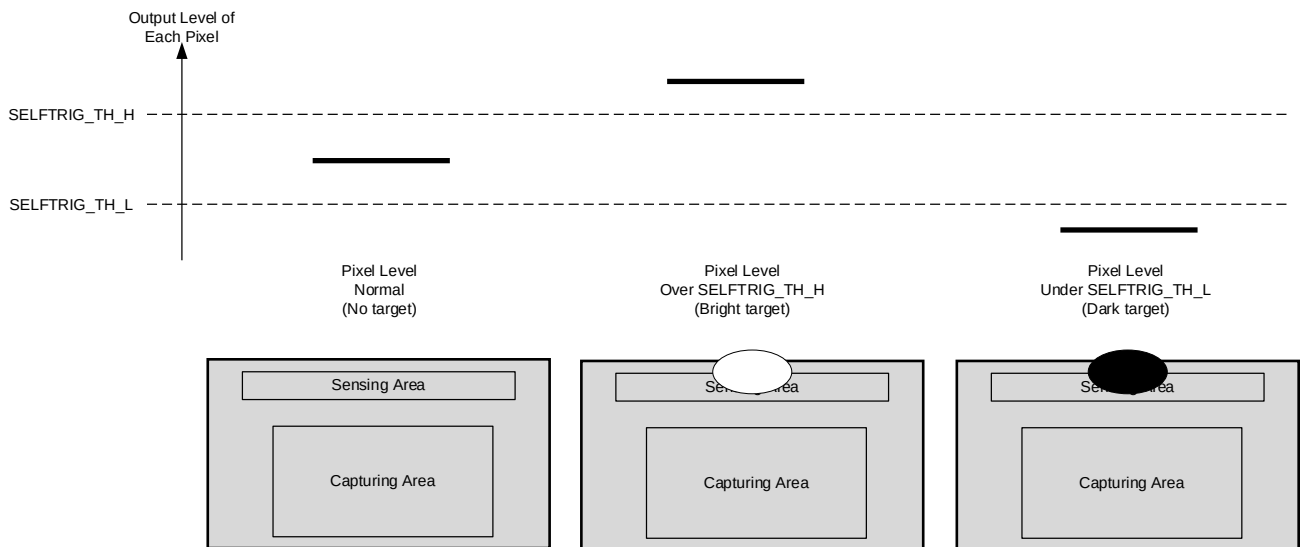


Image of SELFTRIG_TH_H, SELFTRIG_TH_L

SELFTRIG_CNT_TH_H and SELFTRIG_CNT_TH_L are threshold of the number of pixels which are over Level threshold in Sensing Area, and judge to capture the image or not. In case over one of two thresholds, capturing is started.

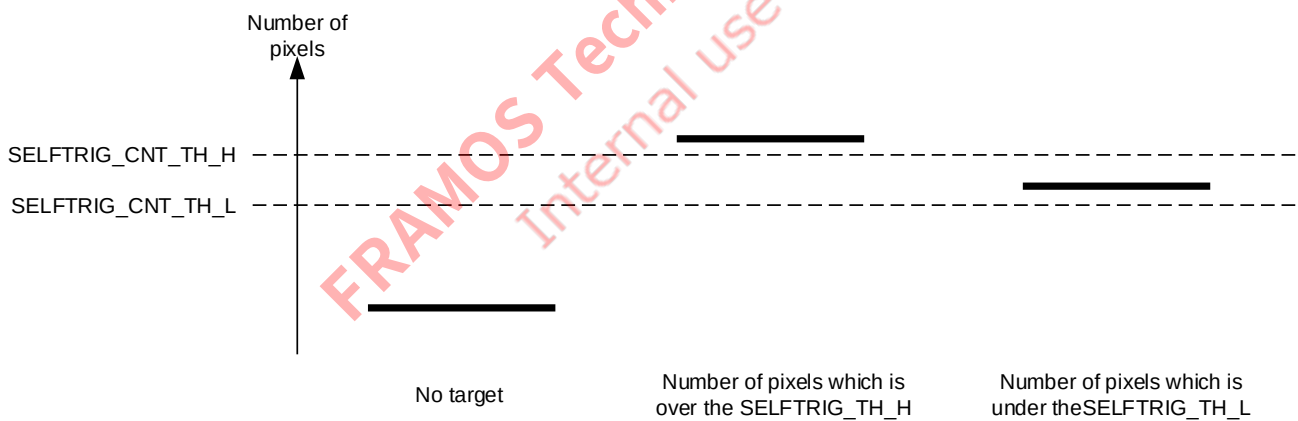
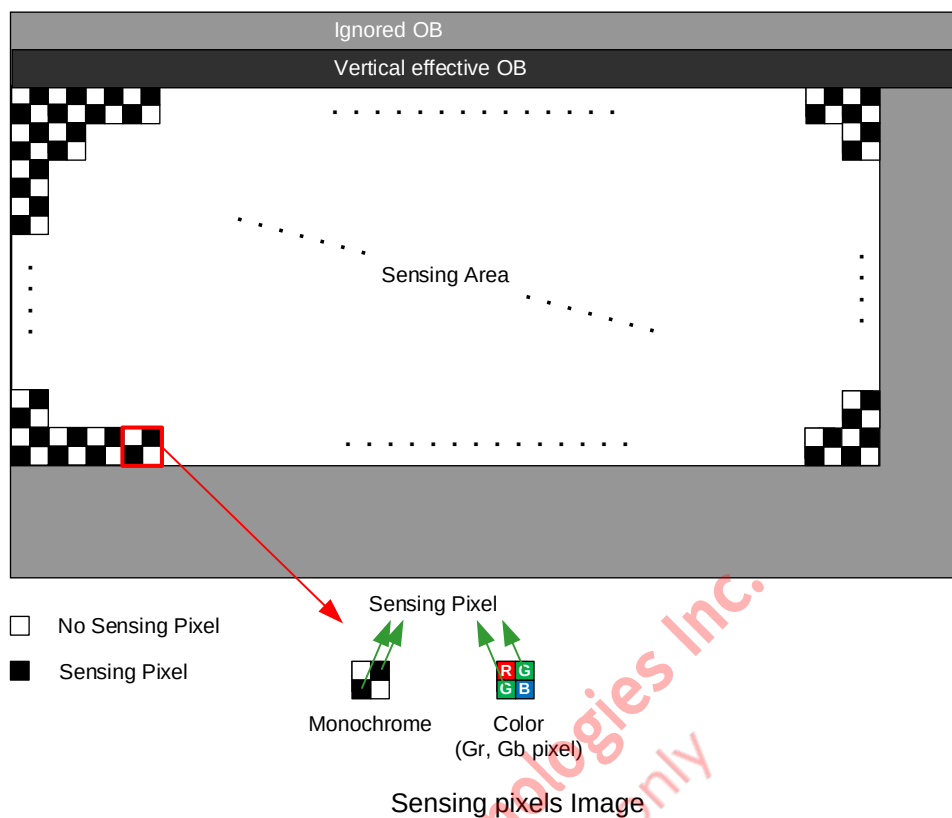


Image of SELFTRIG_CNT_TH_H, SELFTRIG_CNT_TH_L

All pixels in Sensing Area isn't counted, the pixel in Sensing Area as below is counted.



8. Supporting Function of Setting Threshold

This section describes the function which supports thresholds setting.

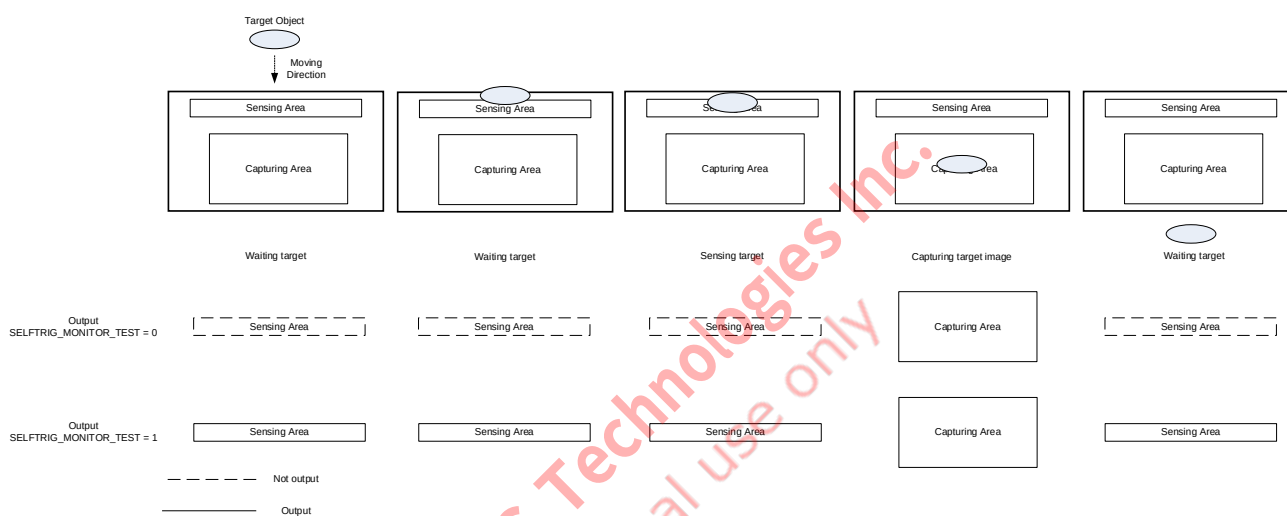
8.1. Sensing Frame Output Setting

By setting the register as below, Sensing frame can be output.

This is to decide the size of each area, position, Level threshold and so on.

By using the initial value of this register, Sensing frame isn't output.

Chip ID	Address	Bit	register name	Note
06h	24h	[0]	SELFTRIG_MONITOR_TEST	Sensing frame output setting 0: Without output 1: With output



Output Image of Setting SELFTRIG_MONITOR_TEST

8.2. How to Confirm Number of Count

The number of detected pixel count of H side or L side in each frame condition is set into the register SELFTRIG_CNT_H (Chip ID = 06h, Address = 28h, 29h, 2Ah), SELFTRIG_CNT_L (Chip ID = 06h, Address = 2Ch, 2Dh, 2Eh).

Confirming this value is used to decide Count Threshold.

The timing to confirm is described on section of Exposure timing and Output timing.

8.3. Self Trigger Forcible Switching Function

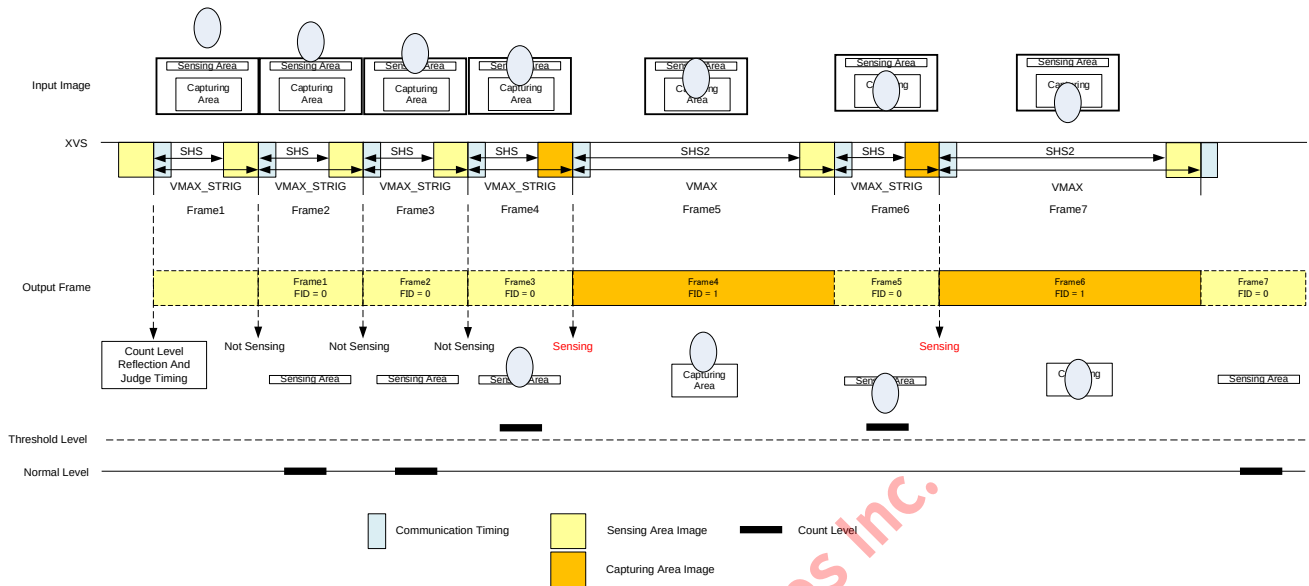
This function is forcible switching Sensing frame and Capturing frame.

By using with Sensing Frame Output Setting, the condition of Sensing frame and Capturing frame can be check.

Chip ID	Address	Bit	register name	Note
06h	24h	[1]	SELF_TRG_FORCE_ON	Self Trigger forcible swiching 0: Normal mode 1: Forced mode (Sensing frame and Capturing frame is switched by each frame.)

9. Exposure Timing and Output Timing

Exposure timing and Output timing is shown as below.



Exposure Timing and Output Timing Image

There is note as below.

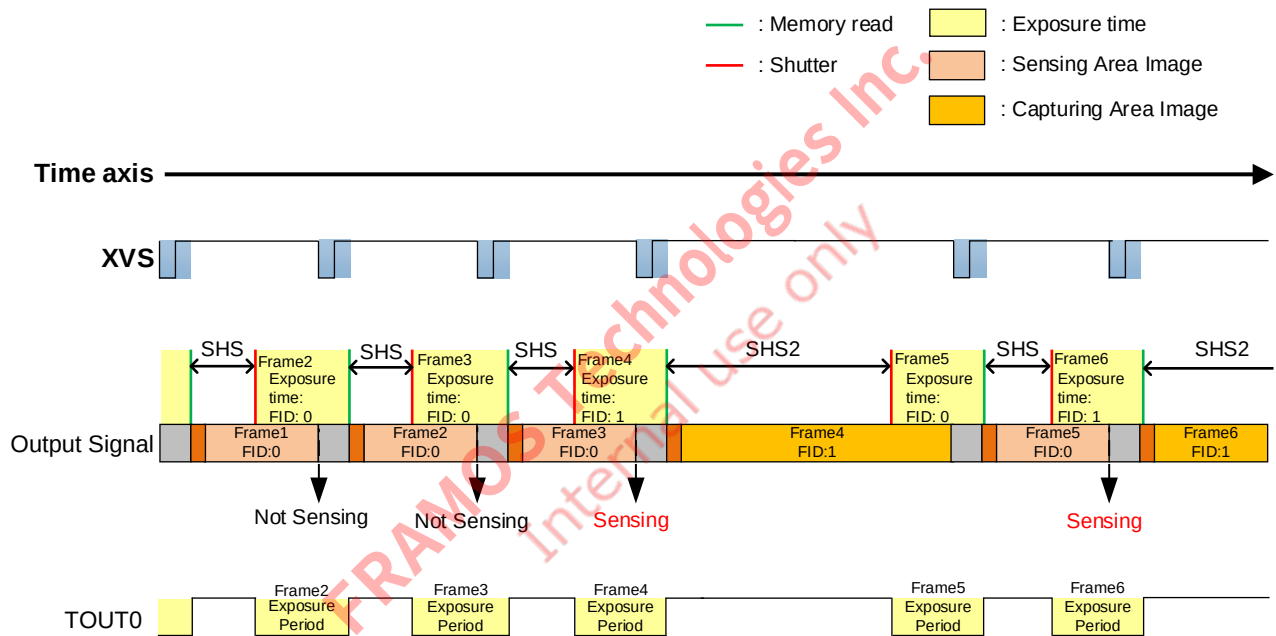
- *1 Exposure time setting has restriction as shown below.
 $VMAX_STRIG - SHS = VMAX - SHS2$
- *2 In case of target in Sensing Area, there is one output of Sensing Area after output of Capturing Area.
- *3 Number of count can be read at communication period of next frame after the output.
- *4 Number of count of Capturing Area is 0.

10. The monitor output for Exposure period

As for this sensor, TOUT0 can output the status of the exposure control signal in Self trigger.

10.1. List of registers

Register	Register details			Initial value	Description
	Chip-ID	Address (I ² C)	bit		
TOUT0SEL [2:0]	06h	36h (3436)	[2:0]	0h	TOUT0 pin setting 0h : Low fixed 1h : Exposure period monitoring



The monitor output for Exposure period of Self Trigger mode

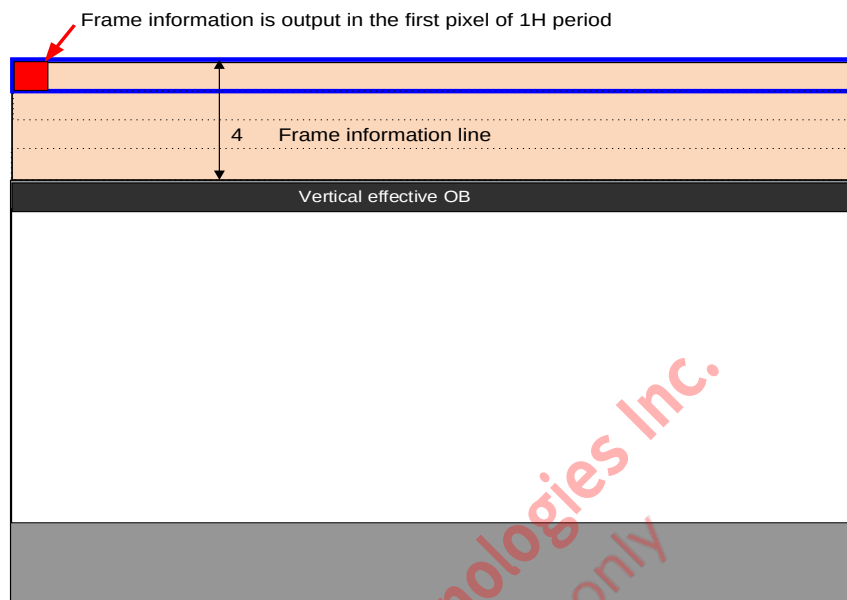
11. Frame Information

The identification signal of each frame is Frame ID.

11.1. Frame information line

This product uses Frame information line of SLVS to output Frame ID.

It is added to the output signal of the 1st pixel of the blue frame in the figure below.



Frame information line output timing

11.2. Output format (Frame information line)

The output format of Frame information line is shown below.

Output	8bit
[7]	1
[6]	0
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	0
[1]	Frame ID [1]
[0]	Frame ID [0]

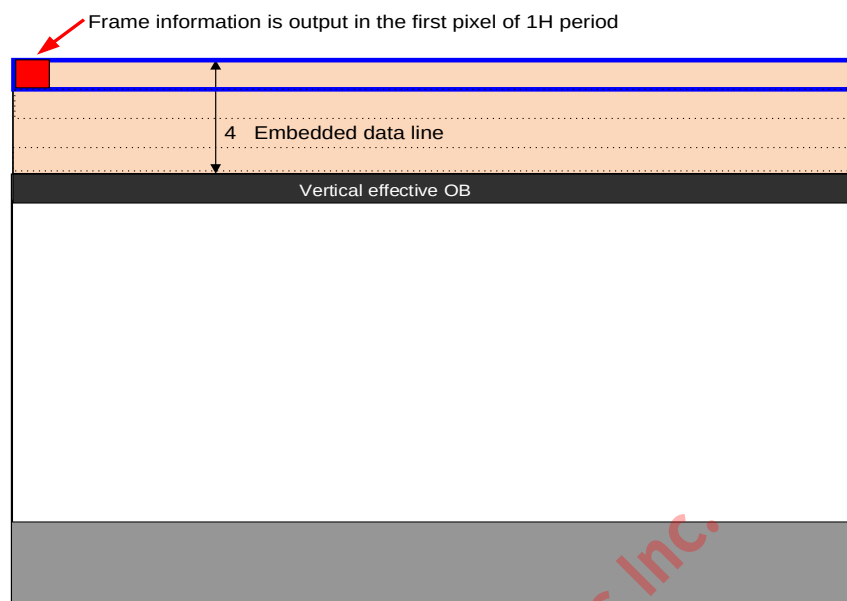
Output	10bit
[9]	1
[8]	0
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	0
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	12bit
[11]	0
[10]	0
[9]	1
[8]	0
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	0
[1]	Frame ID [1]
[0]	Frame ID [0]

Output format (Frame information line)

11.3. Embedded data line

This product uses Embedded data line of SLVS-EC or CSI-2 to output Frame ID. It is added to the output signal of the 1st pixel of the blue frame in the figure below.



Frame information lien output timing

11.4. Output format (Embedded data line)

The output format of Embedded data line is shown below.

(SLVS-EC)

Output	8bit
[7]	1
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	12bit
[11]	0
[10]	0
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output format (Embedded data line (SLVS-EC))

(CSI-2)

Output	8bit
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	Frame ID [1]
[8]	Frame ID [0]
[7]	0
[6]	0
[5]	0
[4]	Frameset
[3]	Frame ID [1]
[2]	Frame ID [0]
[1]	0
[0]	1

Output	12bit
[11]	Frame ID [1]
[10]	Frame ID [0]
[9]	0
[8]	0
[7]	0
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	1
[1]	0
[0]	1

Output format (Embedded data line (CSI-2))

12. Mode Transitions

The transition related to Self Trigger mode has to set via the sensor standby. If perform the transition which is not described in table below, set via sensor standby.

12.1. Mode Transition

Mode Transition		Remark
Normal mode	⇔ Normal mode (Self Trigger mode)	Set via the sensor standby.
Change ROI cropping position and cropping size		If change the size of either the sensing frame or the capture ring frame, please set via the sensor standby.
Normal mode (Self Trigger mode)	⇔ Normal mode (Self Trigger mode)	

FRAMOS Technologies Inc.
Internal use only

13. Note

- Self-trigger mode only supports sensor master mode (normal mode).
* Trigger mode is not supported
- Set VMAX / VMAX_STRIG following to the following restrictions.
VMAX > VNDMY + GMRWT + GMTWT + GSDLY + Vertical effective OB + FID1_ROIWV1 + OFFSET
VMAX_STRIG > VNDMY + GMRWT + GMTWT + GSDLY + Vertical effective OB + FID0_ROIWV1 + OFFSET
* The table below shows the OFFSET of each Readout drive modes

Unit: [XHS]

	All-pixel scan mode		1/2 Subsampling mode		2 × 2 FD binning mode	
	Monochrome / Color		Monochrome / Color		Monochrome	
OFFSET	17 ^{*1}	9 ^{*2}	9 ^{*1}	5 ^{*2}	9 ^{*1}	5 ^{*2}

^{*1} IMX530 series and IMX535 series.

^{*2} IMX540 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

- Driving with the combination of “1 / 2 Subsampling mode” and “ROI mode” is not supported.
- Driving with the combination of “1 / 2 Subsampling mode” and “Overlap ROI mode” is not supported.

Revision History

Version	Date	Page	Remarks
Rev.0.1	2019 / 5 / 17	-	First Edition
Rev.0.2	2019 / 6 / 20	1, 21	Correction: The function name of the monitor output for exposure time
		21	Delete: "continuation" of the sentence
Rev.1.0	2019 / 12 / 9	-	First Edition (Official version of IMX530 series / IMX540 series)
		1	Added: Chapter "Product type" Added: SLVS output (2ch) in the support interface
Rev.2.0	2020 / 12 / 17	-	Added: IMX535 series / IMX545 series / IMX565 series
		2, 25	Added: CSI-2 serial data output
		4, 5, 6, 7, 8, 9	Added: Default Value after reset of IMX535 series, IMX545 series, IMX565 series
		26	Added: OFFSET Value of IMX535 series, IMX545 series, IMX565 series
Rev.3.0	2021 / 3 / 4	1, 2, 4, 7, 8, 9, 27	Added: IMX548 series / IMX568 series
		2	Added: Vertical / Horizontal 1/2 Subsampling mode and 2 x 2 FD binning mode to Support readout drive mode
		5, 6, 7	Correction: Bit width FID0_ROIPV1 [11:0] → [12:0] FID0_ROI WV1 [11:0] → [12:0] FID1_ROIPV1 [11:0] → [12:0] FID1_ROI WV1 [11:0] → [12:0]
		17	Correction: Register width SELFTRIG_CNT_TH_H [15:0] → [23:0] SELFTRIG_CNT_TH_L [15:0] → [23:0]
		27	Added: Two items about combination drive to "13. Note"